

Polynomial Regression

Contents

1. Overview	2
2. A curved pattern	2
3. Adding powers as new columns	2
4. Choosing the degree	4
5. More than one input	5
6. Summary	7
7. Key terms	7

Where this fits: Pipeline map steps 8 (Model), 9 (Evaluate). See the [Course map](#).



- **Builds on:** Train-test split ([Note 13](#)); Multiple linear regression ([Note 55](#)).
- **Leads to:** Bias-variance trade-off ([Note 62](#)); Regularisation ([Note 63](#)); Logistic regression ([Note 70](#)); Hyperparameter tuning ([Note 81](#)); Bagging ([Note 105](#)); Random forest ([Note 108](#)).
- **Compare with:** Decision surface and boundary ([Note 91](#)); Kernel trick ([Note 95](#)).

1. Overview

Key point: Polynomial regression fits curves by adding powers of the input (x^2 , x^3 , ...) as new columns and then running ordinary linear regression on them.

Linear regression fits a straight line (or flat plane). Real relationships are often curved: the output may first fall and then rise, or grow faster and faster. A straight line misses such patterns.

Polynomial regression handles curves without a new algorithm. It creates new input columns from powers of the existing ones and gives them to plain linear regression. The **degree** is the highest power used.

2. A curved pattern

Key point: On data that follows a parabola, a straight line reaches only R^2 0.38.

The example data has 200 points that follow

$$y = 0.8x^2 + 0.9x + 2 + \text{noise}$$

for x between -3 and 3 : a U-shaped curve with some random scatter. Fitted with ordinary linear regression, the best straight line scores only $R^2 = 0.38$ on the test points. It cannot bend, so it misses the low middle and both high ends (the red line in Figure 2, left).

3. Adding powers as new columns

Key point: For degree 2, each row gets a new column holding x^2 . Linear regression then finds one coefficient per column.

The trick: for each row, compute x^2 and add it as a new input column (Figure 1).

x	PolynomialFeatures →	1	x	x^2
-0.89		1	-0.89	0.79
1.50		1	1.50	2.25
2.10		1	2.10	4.41

a new column x^2 (and a column of 1s); the model stays linear in its coefficients: $\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$

Figure 1: PolynomialFeatures turns one column into 1, x and x²

With the columns x and x^2 , linear regression fits

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$

This is a curve in x , but it is still a straight-line combination of its coefficients: each coefficient just multiplies a column. So the usual linear regression machinery (OLS or gradient descent) finds β_0 , β_1 and β_2 unchanged.

That is why polynomial regression is called “linear”: **linear** refers to the coefficients, not to the shape of the curve in x .

Python: Degree-2 polynomial regression.

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

poly = PolynomialFeatures(degree=2, include_bias=True)
X_train_p = poly.fit_transform(X_train)    # [x] -> [1, x, x^2]
X_test_p = poly.transform(X_test)

lr = LinearRegression().fit(X_train_p, y_train)
lr.score(X_test_p, y_test)                # R2 0.83
lr.coef_, lr.intercept_                   # [0, 1.04, 0.82], 1.92
include_bias=True adds the column of 1s; LinearRegression fits its own intercept
anyway, so that column gets coefficient 0 and either setting works.
```

With the x^2 column, the test R^2 jumps from 0.38 to 0.83. The learned equation, $\hat{y} = 1.92 + 1.04x + 0.82x^2$, is close to the true $2 + 0.9x + 0.8x^2$; the remaining difference is the noise.

4. Choosing the degree

Key point: Too low a degree underfits; too high a degree bends to fit the noise and overfits. Training R^2 keeps rising with the degree, but test R^2 falls.

Higher degrees add more columns, x^3 , x^4 and so on, and the curve can bend more. Figure 2 fits three degrees to only 25 training points, then scores them on 200 new test points.

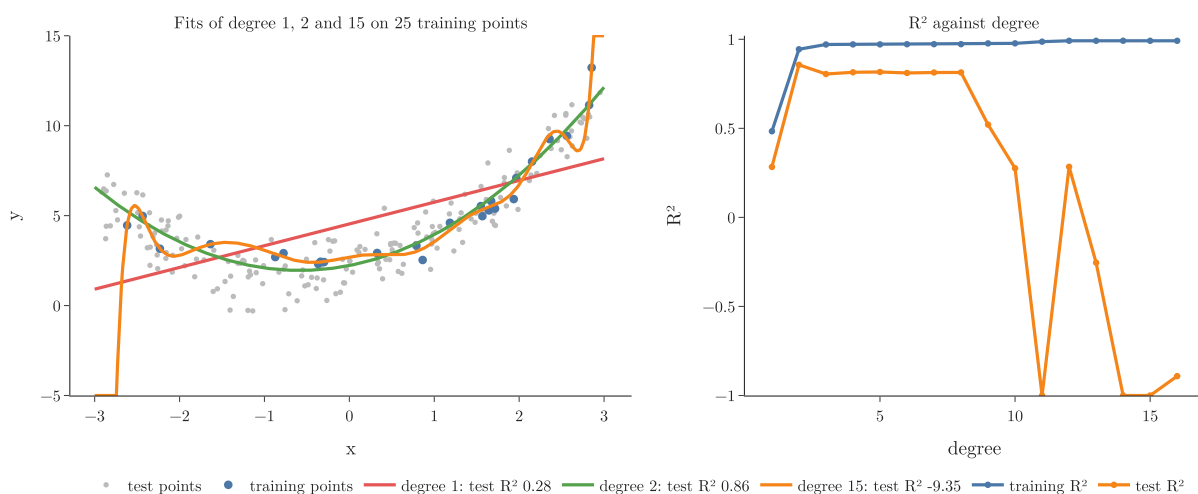


Figure 2: Polynomial fits of degree 1, 2 and 15, and R^2 against the degree

Degree	Training R^2	Test R^2	
1	0.48	0.28	underfits: too simple to follow the curve
2	0.94	0.86	right shape: matches how the data was made
6	0.97	0.81	starting to fit noise
10	0.98	0.28	overfits
15	0.99	-9.35	overfits badly: wild swings between and beyond the points

The right panel shows the pattern. **Training R^2** rises with every extra degree, because more columns always let the curve pass closer to the training points. **Test R^2** peaks at degree 2 and then collapses: the high-degree curve has learned the noise of these 25 points, not the pattern (the challenges and fitting Notes called this overfitting).

The degree is a hyperparameter. It is chosen by comparing scores on data not used for training, as here, or with cross-validation (a later Note).

Extra: A degree-15 polynomial has terms up to x^{15} ; with $x = 3$ that is about 14 million. Columns on such different scales make the fit numerically unstable, which is why high-degree models are usually built as a pipeline of `PolynomialFeatures`, `StandardScaler` and `LinearRegression`, as in Figure 2. The wildness of the degree-15 curve near the edges is the overfitting, not a numerical error.

5. More than one input

Key point: With several inputs, `PolynomialFeatures` also adds products of inputs, such as xy . The number of columns grows very fast with the degree.

With two inputs x and y , degree 2 creates all terms of total power up to 2:

$$1, x, y, x^2, xy, y^2$$

The product xy is an **interaction term**: it lets the effect of x depend on the value of y . The fitted model is a curved surface instead of a flat plane.

Figure 3 shows data made from $z = x^2 + y^2 + 0.2x + 0.2y + 0.1xy + 2$ plus noise. A plane reaches $R^2 = 0.61$; a degree-2 surface reaches 0.98.

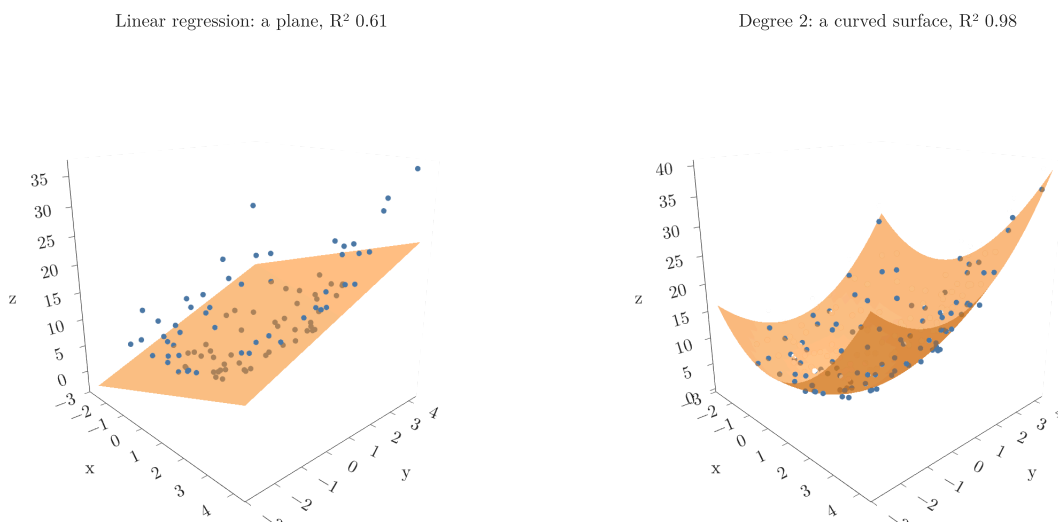


Figure 3: A plane vs a degree-2 surface on curved 3D data

The number of new columns grows quickly. With 2 inputs, degree 2 gives 6 columns; degree 30 gives 496. With many inputs and a high degree, the columns quickly outnumber the rows, and the model overfits.

Extra: To put two 1-D arrays side by side as columns, use `np.c_[x, y]` or `np.column_stack([x, y])`. A shortcut sometimes seen, `np.array([x,`

`y]).reshape(100, 2)`, does not do this: it fills the rows with 100 values of `x` first and then of `y`, so most rows pair unrelated numbers. The model still runs, but on scrambled data. Checking `X[:3]` against `x[:3]` and `y[:3]` catches it.

6. Summary

Degree	Columns for one input	Behaviour on the example
1	x	straight line, underfits (test R^2 0.28)
2	x, x^2	follows the curve (test R^2 0.86)
15	x to x^{15}	overfits (test R^2 -9.35)

- Polynomial regression = new power columns + ordinary linear regression.
- It is “linear” in the coefficients, so OLS and gradient descent work unchanged.
- `PolynomialFeatures(degree=d)` creates the columns; with several inputs it adds interaction terms too.
- The degree controls flexibility: too low underfits, too high overfits. Choose it on held-out data.

7. Key terms

Term	Meaning
Polynomial regression	Linear regression on powers (and products) of the inputs, to fit curves
Degree	The highest power used in the polynomial
<code>PolynomialFeatures</code>	scikit-learn transformer that creates the power and product columns
<code>include_bias</code>	<code>PolynomialFeatures</code> setting that adds a column of 1s
Interaction term	A product of two inputs, such as xy , that lets one input’s effect depend on another
Underfitting	A model too simple to capture the pattern; poor on training and test data
Overfitting	A model so flexible that it learns the noise; good on training data, poor on test data